AWANTIKA SRIVASTAVA

AI Engineer – Embedded & Edge AI | Computer Vision | Linux

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PROFILE SUMMARY

AI Engineer with hands-on experience in Computer Vision, Edge AI deployment, and ML model training. Experienced in working across the AI lifecycle including data collection, labeling, preprocessing, model training, optimization, and deployment on Linux-based systems. Strong understanding of embedded systems fundamentals and exposure to NVIDIA Jetson and edge-based AI pipelines.

CORE TECHNICAL SKILLS

- **Programming & Data Science:** Python, C++, SQL., Pandas, Numpy, Scikit-learn, SciPy
- **Statistics & Mathematics:** EDA, Statistical Modeling, Descriptive Statistics, Hypothesis Testing, Probability, Sampling, A/B Testing
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, LightGBM, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).
- **Deep Learning:** ANN, CNN, RNN, LSTM, BERT, Transfer Learning, Self-Attention, multi-head Attention, Time-Series Forecasting, Tensorflow, PyTorch, Keras
- **NLP:** Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, TextBlob, SpaCy
- **Generative AI & LLMs:** LLMs, Prompt Engineering, RAG, RAG Pipeline, Vector Search, FAISS, Embeddings, Confidence Grading, LangChain
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Paddle OCR, Video Analytics (RTSP), OpenCV
- **Embedded & Electronics:** Digital Electronics, Basic Communication Systems, Microcontroller Fundamentals, ESP32 (Arduino basics – academic exposure), Hardware-Software Integration
- **Edge AI & Deployment:** NVIDIA Jetson (Exposure), Raspberry Pi (Exposure), ONNX, TensorRT (Exposure), Edge Model Optimization
- **MLOps & Tools:** Docker (Basic), Git, CI/CD Basics, Model Versioning
- **Tools & Version Control:** Git, GitHub, Jupyter, Streamlit, PowerBI

EXPERIENCE

AI/ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Designed and implemented end-to-end ML pipelines using **PySpark and Spark SQL** for large structured datasets, enabling scalable model training and inference.
- Built and optimized classical **ML models (XGBoost, Random Forest**, GLMs) improving **prediction** accuracy by **15–20%** across business use cases.
- Deployed scalable ML workflows on **Databricks and AWS**, ensuring modular, production-ready architecture.
- **Developed real-time inference APIs** integrating ML models into applications with **low-latency response**.
- Implemented **robust feature engineering** strategies aligned with complex business objectives.
- Managed experiment **tracking** and model lifecycle using **MLflow** and **Git**-based version control.
- Built **LLM-based automation** tools leveraging **RAG pipelines** and **prompt engineering** for enterprise use cases.
- Applied **SHAP-based model explainability** techniques to improve transparency and stakeholder trust.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed and deployed a** real-time computer vision–based ML system **for unsafe** driver behavior detection using **CNN-based SSD MobileNet** model.
- Trained and optimized models on large-scale video datasets, achieving **20–25 FPS** real-time processing with **<150 ms inference latency**.
- Implemented **end-to-end ML pipelines** for data ingestion, preprocessing, model training, evaluation, and production inference.
- Deployed optimized models using **TensorFlow Lite** on edge/production environments for continuous monitoring.
- Built a **Flask-based web dashboard** to visualize detections and automatically record **30-second event clips**, reducing manual review effort.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Developed **LSTM-based time-series forecasting** model trained on 5+ years historical stock data reducing **RMSE** by 18**%.**
- Engineered lag **features**, **rolling statistics**, and volatility indicators improving prediction stability.
- **Implemented** custom **BPTT training** loop with adaptive learning rate scheduling for **stable convergence**.
- Applied **regularization techniques** reducing overfitting gap by **20**%.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology (Electrical and electronics engineering)